

Lance Nguyen

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EDUCATION

George Mason University – Volgenau School of Engineering

Fairfax, VA

Bachelor of Science in Electrical Engineering

2019-2023

Relevant Coursework: Embedded System Hardware Interfaces, Physics 3, System/Control Theory, Computer Networking, Computer Organization, Linear Electronics II, Digital System Design II, Electrical Circuit Analysis II, Communication Theory

Personal Portfolio (Relevant Projects/Work Experience) - lance-portfolio.lnguyen9152.workers.dev

WORK EXPERIENCE

Relativity Space

Los Angeles, CA

Ground Support Equipment Engineer II – Avionics Test

2026-Present

- Lead engineer for the design, integration, and ongoing development of high-reliability ground support systems, ensuring mission readiness and supporting Terran R's integration, test, and launch phases.
- Design and deliver Engine Checkout Racks to the propulsion team, ensuring robust testing and validation of flight engines with <1 us accuracy.
- Support design of ground support racks for low and high voltage electrical buses used to interface with flight hardware.
- Collaborate with avionics, propulsion, software, manufacturing, and integration teams to ensure reliable operation for vehicle test and launch.
- Design harnesses and junction boxes for electrical power distribution and signal transmission, ensuring seamless connections between ground systems and the vehicle.
- Oversee fabrication, commissioning, and validation of GSE systems, including electrical testing and calibration.
- Integrate electrical systems using protocols such as Ethernet, Modbus, and SCPI for reliable data transfer and control.

Data and Controls Engineer II – Factory Test

2025-2026

- Develop test infrastructure, checkout cabinets, instrumentation boxes, and data systems for hardware verification ensuring 0.01% accuracy.
- Integrate instrumentation (pressure transducers, RTDs, thermocouples, accelerometers, actuators, etc.) into test systems, including channel mapping and signal scaling/conditioning.
- Design test systems and create work instructions using tools such as NX, Altium, and WireViz to document and communicate assembly and integration procedures with a 100% on time completion and reliability rate.
- Develop PLC automation using a proprietary company framework and EtherCAT/Modbus.
- Manage multiple projects simultaneously, became a subject matter expert on DACs backend, and mentor engineers on specialized topics.

British Petroleum

Houston, TX

Instrumentation, Controls, and Electrical Engineer – Production and Operations

2023-2025

- Lead multi-disciplinary teams and contractors to execute high-budget projects, achieving a 95% on-time completion rate and delivering 80% under budget.
- Optimize alarm management, implementing dynamic alarm suppression and reducing high/urgent alarm KPIs by 50%.
- Enhance plant reliability by conducting RCFA (Root Cause Failure Analysis) studies, performing risk assessments, and implementing mitigation strategies, reducing deficiencies by 80%.
- Conduct FAT (Factory Acceptance Test) and SAT (Site Acceptance Test) tests on brownfield projects, ensuring system functionality, integration, and compliance with design requirements.
- Maintain and troubleshoot Rockwell, Honeywell, and Allen-Bradley PLCs, tracing I/O, implementing control loops, and redlining documents for accuracy.
- Design and develop custom HMI screens and faceplates to improve usability and efficiency for offshore operators.

TradersAI

Fairfax, VA

Software Developer

2021-2022

- Assisted in developing an algorithm-based trading strategy for /ES futures, contributing to a 351% return since 2018.
- Designed and built a mobile app to send real-time trading signals using OneSignal for notifications, BeautifulSoup for web data extraction, and Pandas & NumPy for data processing.
- Built Python-based automation for data processing, real-time market retrieval, and trade signal execution.
- Developed an automated system to generate and send real-time trade alerts based on market data.
- Created a cross-platform mobile interface using Figma and AirTable, ensuring a seamless user experience.

ADDITIONAL INFORMATION

Programming Languages: Java, Python, MatLab, Simulink, Verilog, Assembly, Ladder Logic, Functional Block, Structured Text

Technical Skills: Autodesk AutoCAD, P&ID/Electrical/Logic Diagrams, Project Management, Gantt Charts, PSpice, Osmond PCB, Modbus/Ethernet/DeviceNet/ControlNet, Networking, Factory/Site Acceptance Testing, Document Control, Linux Command Line, VMWare Virtualization, Risk Assessments, Execution Planning, Budget Planning